Engineering continuous measurements for measurement-based holonomic quantum computation

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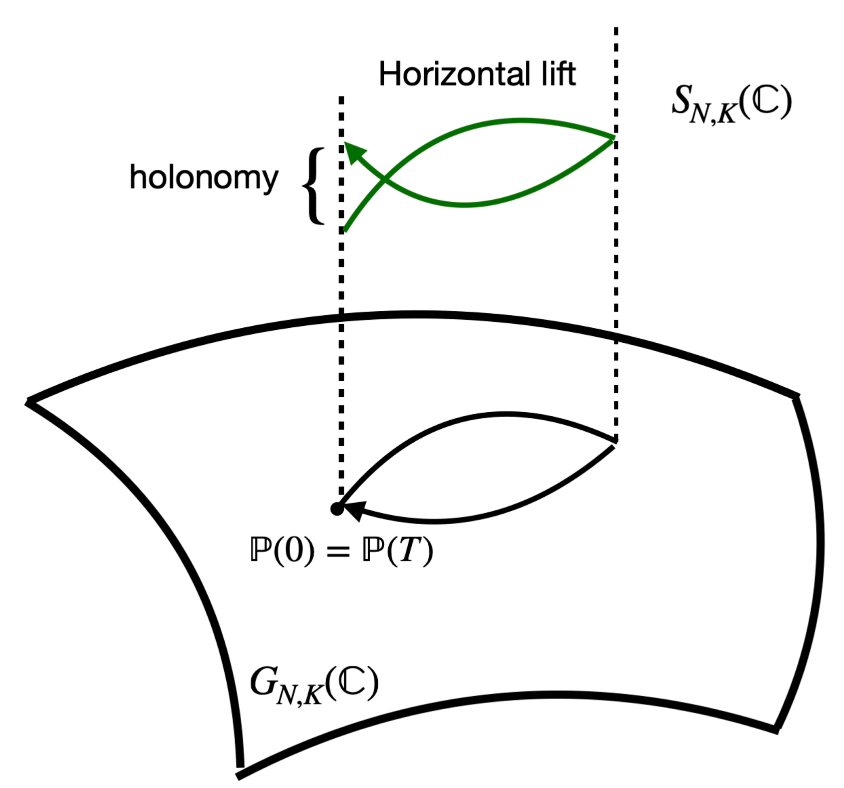
Introduction

Measurement-based holonomic quantum computation (MHQC) enables robust logical operations on quantum stabilizer codes via holonomies, continuous measurements, and the quantum Zeno effect. Continuous measurement of slowly-changing stabilizers both corrects Markovian errors and suppresses non-Markovian noise. A major challenge is implementing complex multi-qubit measurements on current hardware. We propose an indirect method using few ancillas, pairwise interactions, and fast single-qubit pulses, enabling a universal logical gate set for small codes and making MHQC feasible on near-term quantum devices.

Measurement-based Holonomic Quantum Computation (MHQC)

- In this scheme, a logical unitary gate is applied via generating a holonomy in the following manner:



where $G_{N,K}(\mathbb{C})$ is the complex Grassmannian manifold, $S_{N,K}(\mathbb{C})$ is the fiber bundle called Stiefel manifold.

$$G_{N,K}(\mathbb{C}) = \{\mathbb{P} \in M(N,N;\mathbb{C}) | \mathbb{P}^2 = \mathbb{P}, \mathbb{P}^\dagger = \mathbb{P}, \mathrm{Tr}\{\mathbb{P}\} = k\}, \tag{1}$$

$$S_{N,K}(\mathbb{C}) = \{L \in M(N,K;\mathbb{C}) | L^\dagger L = I_K\}. \tag{2}$$

- A point $\mathbb{P} \in G_{N,K}(\mathbb{C})$ represents the projector onto the code space, while a point, say $L \in S_{N,K}(\mathbb{C})$, in the fiber over $\mathbb{P}$ represents the code basis states.
- We move the point $\mathbb{P}$ along a curve on the Grassmannian manifold using a parameterized unitary operator $V(t)$, such that at $t = T$, the curve forms a loop and we are back to our original point.

$$\mathbb{P}(t) = V(t)\mathbb{P}(0)V^\dagger(t)\,. \tag{3}$$

- This induces a horizontal lift, which produces the desired holonomy, which is the unitary operation we wished to apply. The figure above captures this.
- In the context of QECC, consider an $[\![n,k,d]\!]$ stabilizer code with stabilizer generators $\langle S\rangle = \{g_i\}_{i=1}^{n-k}$. If we wish to apply the logical gate

$$\overline{U} = \exp(i\theta H)\,. \tag{4}$$

- The unitary we choose to adiabatically change the code by rotating it, is

$$V(t) = \exp\left(i\frac{\theta\omega}{2\pi}H\right)\exp(i\omega t R). \tag{5}$$

- We continuously measure the rotated stabilizers

$$\langle S(t)\rangle = \{V(t) g_i V^\dagger(t)\} = \{g_i(t)\} \tag{6}$$

with measurement rate κ, over the interval $t \in [0, 2\pi/\omega]$ and achieve the desired logical unitary $\overline{U}$ applied to our codespace. However, measuring these $\{g_i(t)\}$ is difficult.

Indirectly measuring $g(t)$

- We implement indirect measurement of $g(t)$ using a single ancilla qubit m initialized in state $|0\rangle$. We apply a time dependent Hamiltonian

$$H(t) = \lambda(\mathbb{P}\_(t) \otimes X_m), \tag{7}$$

where $\mathbb{P}\_(t)$ projects onto -1 eigenspace $g(t)$ and we continously monitor the ancilla qubit m in the Z basis. The dynamics are given by the following stochastic Schrödinger's equation.

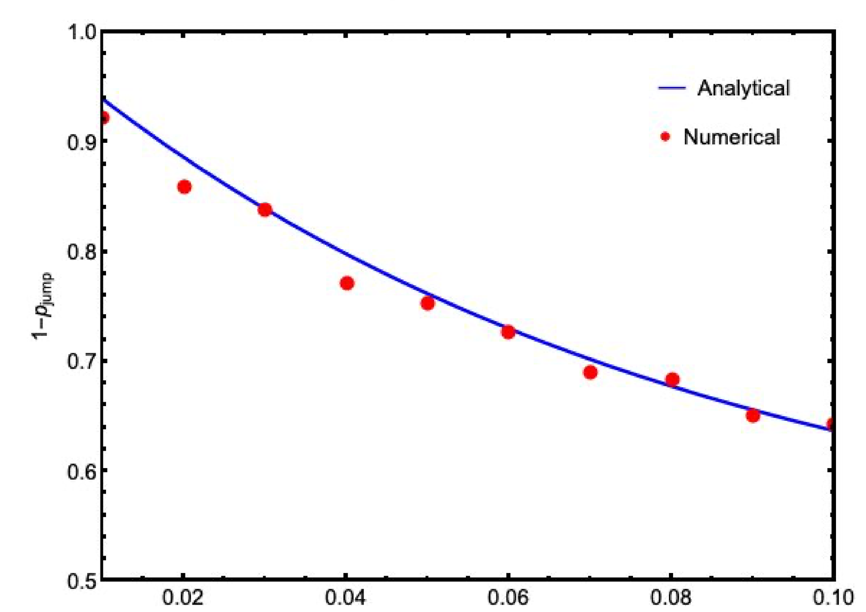
$$|\mathrm{d}\psi\rangle = -iH(t)\,|\psi\rangle\,\mathrm{d}t - \frac{\gamma}{2}\left(Z_m - \langle Z_m|Z_m\rangle\right)^2|\psi\rangle\,\mathrm{d}t + \sqrt{\gamma}\left(Z_m - \langle Z_m|Z_m\rangle\right)|\psi\rangle\,\mathrm{d}W_t\,, \tag{8}$$

where γ is the measurement rate of the ancilla. The relation between this and the effective measurement rate of $g(t)$ is $\kappa = \lambda^2 4\gamma$.

- While doing these measurements, there is a small probability of jumping into the error space defined by the operator R, which we used rotate the stabilizers. The probability of no jump is

$$1 - p_{\text{jump}} = \frac{1}{2}\left[1 + \left(1 - \frac{2\gamma\omega\theta^2}{\pi\lambda^2}\right)\exp\left(-\frac{16\gamma\pi\omega}{\lambda^2}\right)\right]. \tag{9}$$

- By doing the numerical simulations, we verified the equation above and observe that probablity of no jump decreases with increase in adiabaticity.



Constructing Hamiltonian $H(t)$ with pairwise interactions for $[\![4,1,2]\!]$

- Consider the code with the following stabilizer generators

$$g_1 = X_1X_2X_3X_4\,,\quad g_2 = Z_1Z_2Z_3Z_4\,,\quad g_3 = Z_1Z_2. \tag{10}$$

- We aim to implement the logical gate $\overline{U} = R_{\overline{X}}(\theta) = \exp(i\theta H)$, where $H = X_1X_2$. We select a rotation operator $R = Y_1Y_3$. We can observe that during the codespace rotation, g_1 and g_2 remain unchanged. The last stabilizer have the form

$$g_3(t) = \cos(2\omega t)Z_1Z_2 - \sin(2\omega t)\sin\left(\frac{\theta}{\pi}\omega t\right)Y_2Y_3 - \sin(2\omega t)\cos\left(\frac{\theta}{\pi}\omega t\right)X_1Z_2Y_3. \tag{11}$$

- The Hamiltonian we need to apply to measure $g_3(t)$, given in Eq.(7), gives the following unitary evolution for a small timestep δt

$$U(t+\delta t, t) = \exp(-iH(t)\delta t) + O(\lambda^2\delta t^2) \tag{12}$$

$$= \exp\left(-i(f_0H_0 + f_1(t)H_1 + f_2(t)H_2 + f_3(t)H_3)\delta t\right) + O(\lambda^2\delta t^2) \tag{13}$$

$$\approx \underbrace{\exp(-if_0\delta t H_0)}_{H_0\text{ term}}\left(\underbrace{\exp\left(-i\frac{f_1(t)\delta t}{n}H_1\right)}_{H_1\text{ term}} \cdot \underbrace{\exp\left(-i\frac{f_2(t)\delta t}{n}H_2\right)}_{H_2\text{ term}} \cdot \underbrace{\exp\left(-i\frac{f_3(t)\delta t}{n}H_3\right)}_{H_3\text{ term}}\right)^n \tag{14}$$

where

$$f_0 = \tfrac{\lambda}{2},\quad f_1(t) = -\tfrac{\lambda}{2}\cos(2\omega t),\quad f_2(t) = \tfrac{\lambda}{2}\sin(2\omega t)\sin\left(\tfrac{\theta}{\pi}\omega t\right),\quad f_3(t) = \tfrac{\lambda}{2}\sin(2\omega t)\cos\left(\tfrac{\theta}{\pi}\omega t\right),$$
$$H_0 = X_5,\quad H_1 = Z_1Z_2X_5,\quad H_2 = Y_2Y_3X_5,\quad H_3 = X_1Z_2Y_3X_5. \tag{15}$$

- To generate H_1, H_2 and H_3 using pairwise interactions, we use the following identities

Lemma 1: For any operators A and B,

$$\Omega_{\delta\tau}(A,B) \equiv e^{iB\delta\tau}e^{iA\delta\tau}e^{-iB\delta\tau}e^{-iA\delta\tau} \tag{16a}$$

$$= e^{-i\delta\tau^2(i[A,B])} + O(\delta\tau^3). \tag{16b}$$

Lemma 2: Define the unitary U

$$U = e^{iC\delta\tau}\Omega_{\delta\tau}(A,-B)e^{-iC\delta\tau}\Omega_{\delta\tau}(A,B). \tag{17}$$

Let u be a sequence of single qubit pulse(s) such that

$$\{u, [[A,B],B]\} = 0. \tag{18}$$

Then

$$UuUu^\dagger = e^{i2\delta\tau^3[[A,B],C]} + O(\delta\tau^4). \tag{19}$$

- A valid choice is as follows:

$\overline{X}$	A		B		$\delta\tau$
	$f_i(t) > 0$	$f_i(t) < 0$	$f_i(t) > 0$	$f_i(t) < 0$	
H_1 term	Y_2X_5	Z_1X_2	Z_1X_2	Y_2X_5	$\sqrt{\frac{\lvert f_1(t)\rvert\,\delta t}{2n}}$
H_2 term	Y_2X_3	Z_3X_5	Z_3X_5	Y_2X_3	$\sqrt{\frac{\lvert f_2(t)\rvert\,\delta t}{2n}}$

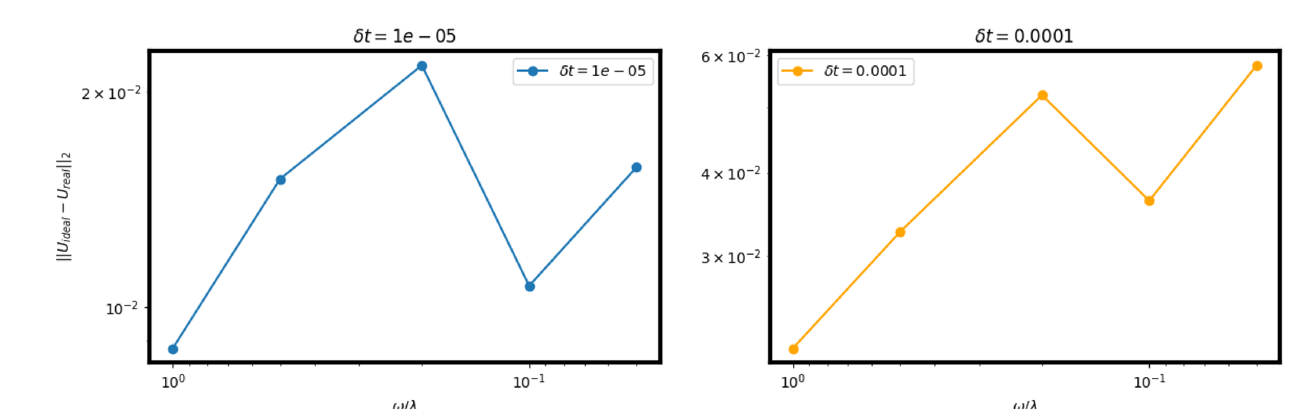
$\overline{X}$	A		B		C	u	$\delta\tau$
	$f_3(t) > 0$	$f_3(t) < 0$	$f_3(t) > 0$	$f_3(t) < 0$			
H_3 term	Y_2X_3	X_1X_2	X_1X_2	Y_2X_3	Z_3X_5	Z_2	$\sqrt[3]{\frac{\lvert f_3(t)\rvert\,\delta t}{8n}}$

Error Analysis

- The total error of generating the required Hamiltonian using pairwise interactions and fast pulses is

$$E_{\text{tot}} = \begin{cases} O\left(\frac{\lambda^{4/3}\delta t^{1/3}}{\omega n^{1/3}}\right) & n = 1 \\ O\left(\frac{\lambda^{4/3}\delta t^{1/3}}{\omega n^{1/3}}\right) & n \geq 2 \text{ and } \lambda\delta t < 1/\sqrt{n} \\ O\left(\frac{\lambda^2\delta t}{\omega}\right) & n \geq 2 \text{ and } \lambda\delta t > 1/\sqrt{n} \end{cases} \tag{20}$$

- The following points provide numerical verification that the error bounds hold:



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- P. Zanardi, & M. Rasetti, *Holonomic quantum computation*, Physics Letters A, 264(2–3), 94–99 (1999).
- M. Marvian, T. A. Brun, and D. A. Lidar, *Suppression of effective noise in hamiltonian simulations*, Physical Review A 96 (2017).

Don't miss the SQuInT 2025 talk!

My co-author Anirudh Lanka is giving a talk on *continuous measurement-based holonomic quantum computation* on **10/10/25**, **5:30-6:00 PM** in Session 09c!

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